



Introduction to Photogrammetry

Lecture 06

Components of Aerial Cameras

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In Today's Class

1. Introduction
2. Main Components of an Aerial Camera
 - A. Lens assembly
 - B. Focal plane
 - C. Inner cone
 - D. Outer cone
 - E. Drive mechanism
 - F. Film magazine
3. Importance of Camera Design

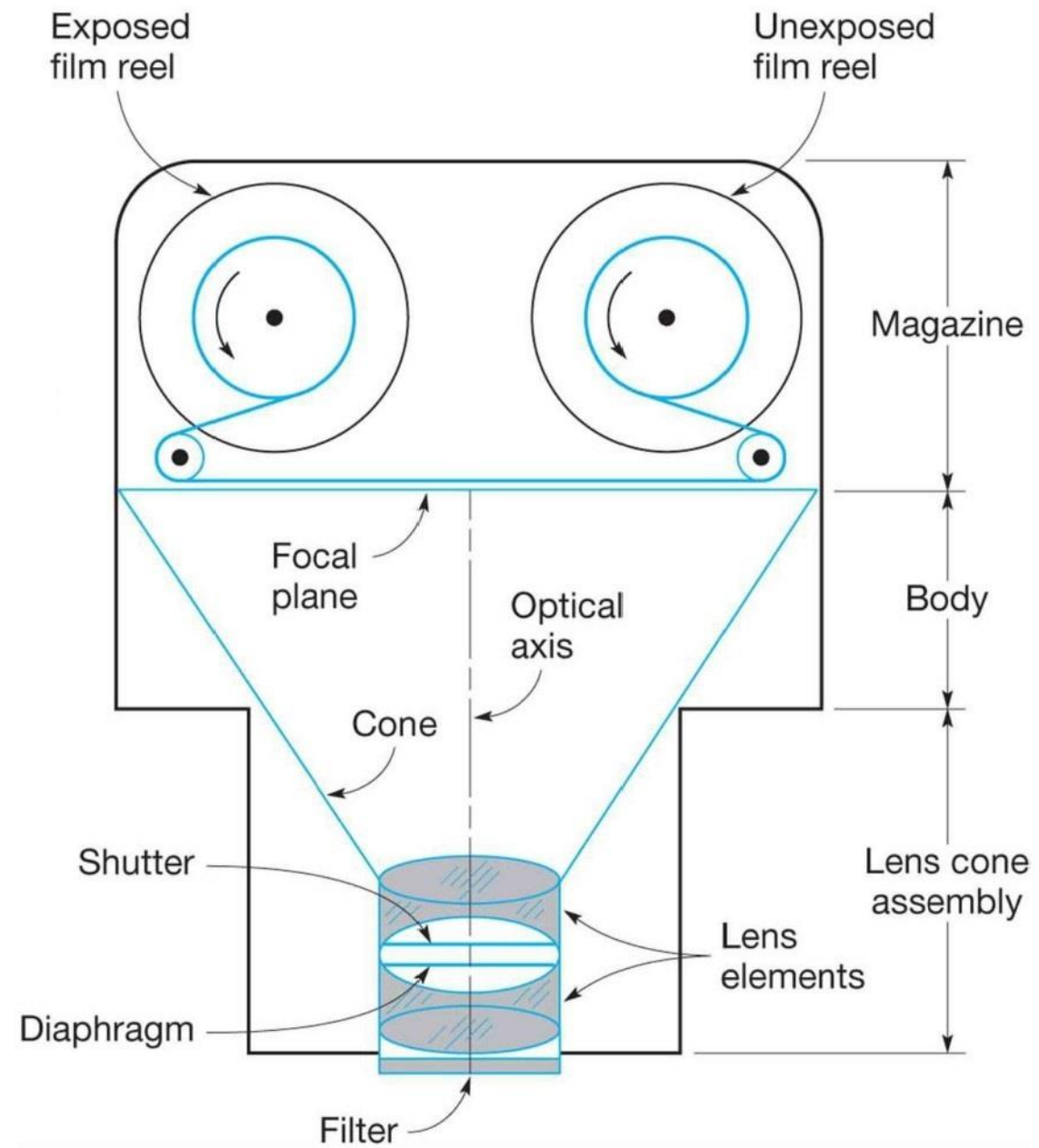
1. Introduction

- Aerial cameras are specially designed for accurate mapping.
- Each component of the camera performs a specific function.
- Proper functioning of these parts ensures **high-quality aerial photographs**.
- Understanding these components helps students interpret aerial images correctly.

- The aerial camera is built as a **strong and stable system**.
- All internal components must remain **precisely aligned**.
- The camera body protects the internal components from vibration and temperature changes.
- This design ensures accurate photogrammetric measurements.

2. Main Components of an Aerial Camera

- A typical aerial camera contains several major parts:
 - A. Lens assembly
 - B. Focal plane
 - C. Inner cone
 - D. Outer cone
 - E. Drive mechanism
 - F. Film magazine
- These components work together to capture aerial photographs.



A. Lens Assembly

- The **lens assembly** is one of the most important parts of the aerial camera.
- Its main function is to focus incoming light onto the film plane.
- The lens assembly also controls the amount of light reaching the film.
- A high-quality lens ensures sharp and clear aerial photographs.

- The lens assembly usually includes:
 - Camera lens (objective lens)
 - Diaphragm (aperture)
 - Shutter
 - Optical filters
- Each of these parts contributes to image quality.

Camera Lens

- The camera lens gathers light reflected from objects on the ground.
- It focuses this light onto the film surface.
- Aerial camera lenses are carefully designed to minimize distortions.
- High optical quality is necessary for accurate photogrammetric measurements.

Diaphragm (Aperture)

- The diaphragm controls the **amount of light** entering the camera.
- It works by adjusting the size of the opening inside the lens.
- A larger opening allows more light to enter.
- A smaller opening reduces the amount of light reaching the film.

Camera Shutter

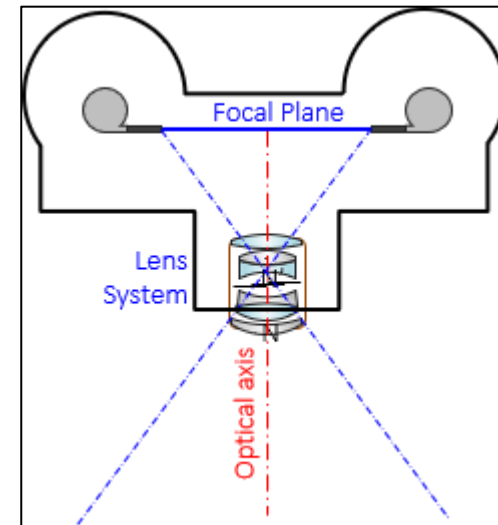
- The shutter controls the **exposure time** of the photograph.
- It opens for a short period to allow light to reach the film.
- After exposure, the shutter closes again.
- Correct exposure time ensures properly illuminated photographs.

Optical Filters

- Optical filters are sometimes used in aerial photography.
- They allow certain wavelengths of light to pass through while blocking others.
- Filters can improve image clarity and contrast.
- They are also useful in environmental and vegetation studies.

B. Focal Plane

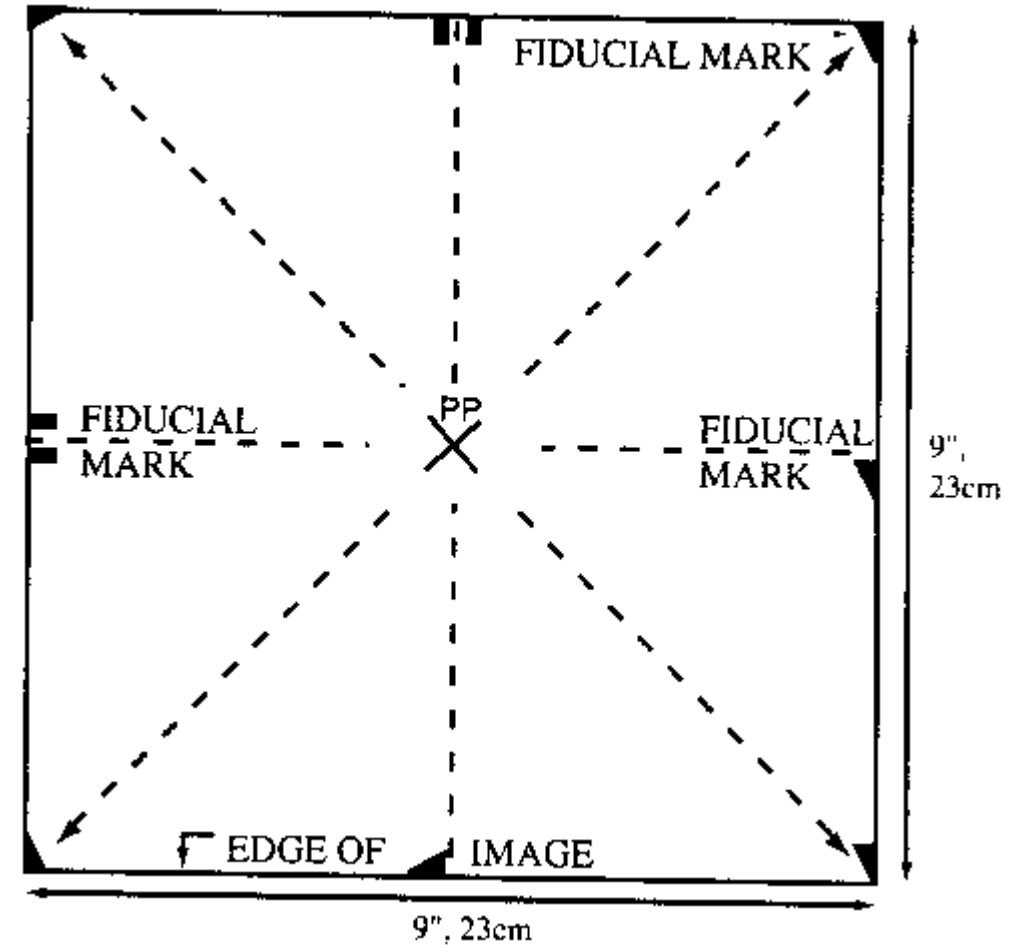
- The **focal plane** is the surface where the image is formed.
- In film cameras, the photographic film is placed on this plane.
- Light focused by the lens creates the image on the film surface.
- Maintaining a perfectly flat focal plane is very important for image accuracy.



Fiducial Marks

- The focal plane contains **small reference marks called fiducial marks**.
- These marks define the coordinate system of the photograph.
- They are used during photogrammetric processing.
- Fiducial marks help determine the geometric orientation of the image.

- Fiducial marks are usually located at the **corners or edges of the photograph**.
- Some cameras have **four fiducial marks**, while others may have **eight**.
- These marks remain fixed relative to the camera lens.
- They are essential for accurate measurements in photogrammetry.



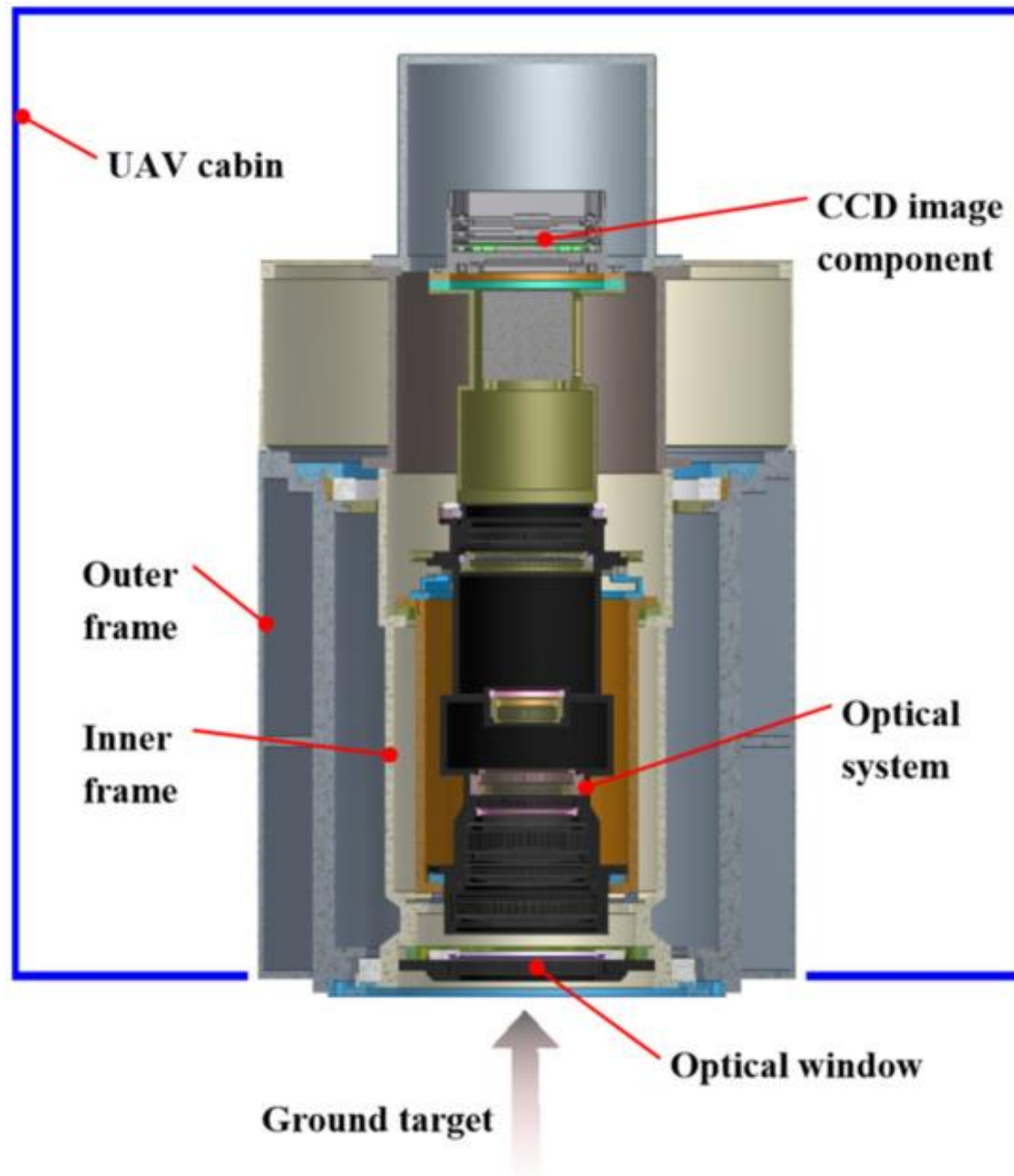
C. Inner Cone

- The **inner cone** connects the lens assembly with the focal plane.
- It maintains the precise distance between the lens and the film.
- The inner cone is usually made from materials with **low thermal expansion**.
- This prevents geometric changes due to temperature variations.

- Even small changes in camera geometry can affect measurements.
- The inner cone keeps the lens and film in a fixed position.
- This stability ensures consistent and reliable photographs.
- Therefore, it plays an important role in photogrammetric accuracy.

D. Outer Cone

- The **outer cone** surrounds and protects the inner cone.
- It forms part of the camera housing.
- The outer cone also supports other mechanical components of the camera.
- It provides structural strength and protection.



E. Drive Mechanism

- The **drive mechanism** controls several mechanical functions of the camera.
- It operates the shutter system.
- It advances the film between exposures.
- It also controls other internal movements of the camera.

- The film must move forward after each photograph is taken.
- The drive mechanism automatically advances the film.
- This prepares the camera for the next exposure.
- Accurate film movement ensures proper spacing between photographs.

F. Film Magazine

- The **film magazine** stores the photographic film.
- It contains both exposed and unexposed film rolls.
- The magazine is detachable so it can be replaced easily during flight missions.
- This allows long aerial photography missions without interruption.

- A vacuum system is used to keep the film **perfectly flat** during exposure.
- Air is removed from behind the film to hold it tightly against the focal plane.
- This prevents film bending or movement.
- A flat film surface ensures sharp and accurate images

3. Importance of Camera Design

- The design of aerial cameras focuses on **precision and stability**.
- Every component is carefully engineered to maintain geometric accuracy.
- High-quality aerial photographs depend on the proper functioning of these components.
- This is why photogrammetric cameras are different from ordinary cameras.



Thank you!

Any Questions
